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Samuele Ferri

Education

- 2024–now **PhD in Computer science and Mathematics of Computation**, *University of Insubria*, Como
Currently writing the thesis. Expected defense date: February 2027
- 2020–2022 **Master degree in mathematics**, *University of Insubria*, Como
Graduation grade: 104/110
- 2022 **Teaching qualification (Italy)**, *University of Insubria*, Como
24 cfu qualification
- 2017–2020 **Bachelor degree in mathematics**, *University of Insubria*, Como
Graduation grade: 93/110
- 2013–2017 **High school diploma**, *Liceo Lugano 1*, Lugano

Publications

Research interests Numerical linear algebra, GLT theory, numerical methods for PDEs

Braun, E. C. and Bretti, G. and Ferri, S. and Santarelli, M. L. and Semplice, M.. "*Dissolution of carbonate stones caused by CO₂ pollutant: an erosion model*". Arxiv preprint (2026)

Ferri, S. and Giraud, C. and Loi, V. and Kuchta, M. and Serra-Capizzano, S.. "*Spectral analysis of the stiffness matrix sequence in the approximated Stokes equation*". Bol. Soc. Mat. Mex, Volume 32, pp. 102 (2026)

Barakitis, N. and Donatelli, M. and Ferri, S. and Loi, V. and Serra-Capizzano, S. and Sormani, R. L.. "*Blocking structures, approximation, and preconditioning*". Numerical Algorithms, Volume 100, pp. 1893–1927 (2025)

Master thesis

Title *De exoticis 7-sphaeris*

Supervisor Giovanni Bazzoni

Description In this paper, we retrace Milnor's studies aimed at finding homeomorphic but not diffeomorphic manifolds on the 7-sphere, through the study of 3-sphere bundles on the 4-sphere and their characteristic classes.

Bachelor thesis

Title *Mixing time for Glauber dynamics*

Supervisor Andrea Posilicano

Description This thesis work is aimed at studying the time required for the Markov chain describing the Glauber dynamics for the Ising model to reach its stationary distribution.

Computer skills

Advanced Python, $\text{\LaTeX}$, C++, Matlab
Intermediate HTML, CSS, Affinity
Basic Javascript, Java, C#, SQL, LabView

Language skills

Mother t. Italian
Advanced English
Basic German, French

Experiences

- 2026-now **Tutoring**, *University of Insubria*, Como
Covering the role of disciplinary tutor for the first year of the bachelor's degree in mathematics.
- 2026 **Internship**, *Università della Svizzera Italiana*, Lugano
From January to July.
- 2025-2026 **Tutoring**, *University of Insubria*, Como
Covered the role of disciplinary tutor for the first year of the bachelor's degree in mathematics.
- 2025 **Internship**, *Institute for applied mathematics "Mauro Picone"*, National Research Council of Italy, Rome
From January to July.
- 2024-2025 **Tutoring**, *University of Insubria*, Como
Covered the role of disciplinary tutor for the first year of the bachelor's degree in mathematics.
- 2024 **Tutoring**, *University of Insubria*, Como
Covered the role of tutor for the "Analisi 2" course of the bachelor's degree in mathematics.
- 2024 **Tutoring**, *University of Insubria*, Como
Covered the role of tutor for the "Matematica 2" course of the bachelor's degree in chemistry.
- 2023 **Data analyst**, *Tools for Smart Minds*, Castel Mella (BS)
From June to December.
- 2023 **Substitution**, *Canobbio Middle School*, Canobbio
Substitute for mathematics for a first grade.
- 2023 **Substitution**, *Centro Professionale di Trevano*, Canobbio
Substitute three teachers for mathematics for vocational baccalaureate classes.
- 2023 **Substitution**, *Liceo Lugano 1*, Lugano
Substitute for two teachers for mathematics and computer science.
- 2022 **Substitution**, *Canobbio Middle School*, Canobbio
Substitute for basic mathematics for a third and a fourth grade.
- 2021-2022 **Tutoring**, *University of Insubria*, Como
Covered the role of tutor for the "Geometria 2" course of the bachelor's degree in mathematics.